

CENTRAL BOARD OF SECONDARY EDUCATION
HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)
SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 2: CELL: THE BUILDING BLOCK OF LIFE

TIME ALLOWED: 45 MINUTES**MAXIMUM MARKS: 25****PAPER CODE: 086/CH02****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Who first observed living, free-moving microscopic cells in pond water using an improved microscope in 1674?**

- (A) Robert Hooke (B) Anton van Leeuwenhoek
(C) Robert Brown (D) Rudolf Virchow

Q.2 Which scientist expanded the Cell Theory by stating 'Omnis cellula-e cellula' (all cells arise from pre-existing cells)?

- (A) Matthias Schleiden (B) Theodor Schwann
(C) Rudolf Virchow (D) Purkinje

Q.3 The plasma membrane is described as selectively permeable primarily because it:

- (A) Allows all substances to enter freely (B) Permits only specific molecules to pass through while preventing others
(C) Is made entirely of impermeable lignin (D) Does not allow water molecules to cross

Q.4 If red blood cells are placed in a hypotonic salt solution, what physical change will occur?

- (A) They will shrink due to exosmosis (B) They will swell and may burst due to endosmosis
(C) They will remain unchanged in size (D) Their cell walls will become turgid

Q.5 The shrinkage of plant cytoplasm away from the cell wall when placed in a concentrated hypertonic sugar solution is called:

- (A) Endosmosis (B) Plasmolysis
(C) De-plasmolysis (D) Imbibition

Q.6 Which cell organelle is responsible for synthesizing lipids and detoxifying poisons and drugs in vertebrate liver cells?

- (A) Rough Endoplasmic Reticulum (RER) (B) Smooth Endoplasmic Reticulum (SER)
(C) Golgi apparatus (D) Ribosome

Q.7 Which organelle consists of a stack of membrane-bound cisternae involved in packaging and dispatching proteins?

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|------------------|---------------------|
| (A) Mitochondria | (B) Golgi apparatus |
| (C) Vacuole | (D) Centrosome |

Q.8 Why are lysosomes known as the 'suicidal bags' of a cell?

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|-----------------------------------|--|
| (A) They produce poisonous toxins | (B) When a cell is damaged, they burst and their powerful digestive enzymes digest the cell itself |
| (C) They starve the cell of ATP | (D) They absorb harmful UV radiation |

Q.9 Which two organelles possess their own DNA and 70S ribosomes, enabling them to make some of their own proteins?

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|---|-----------------------------|
| (A) Mitochondria and Chloroplasts | (B) Lysosomes and Ribosomes |
| (C) Golgi apparatus and Endoplasmic Reticulum | (D) Nucleus and Vacuole |

Q.10 The inner membrane of a mitochondrion is deeply folded into cristae. What is the functional advantage of these folds?

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|---|---|
| (A) To store extra water | (B) To vastly increase surface area for ATP-generating chemical reactions |
| (C) To protect the mitochondrial DNA from radiation | (D) To give structural rigidity to the organelle |

Q.11 A colourless plastid responsible for storing starch, oils, or protein granules in plant cells is termed:

- | | |
|-----------------|---------------------|
| (A) Chromoplast | (B) Chloroplast |
| (C) Leucoplast | (D) Amyloplast only |

Q.12 The cell wall of plants is primarily composed of which complex carbohydrate that provides structural strength?

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|--------------|---------------|
| (A) Glycogen | (B) Cellulose |
| (C) Chitin | (D) Starch |

Q.13 In a mature plant cell, turgidity and rigidity are maintained mainly by:

- | | |
|----------------------------------|--|
| (A) Centrioles | (B) A large central sap vacuole filled with cell sap |
| (C) Lysosomal hydrolytic enzymes | (D) Chromatin fibres |

Q.14 Which type of cell division leads to the formation of two genetically identical diploid daughter cells for body growth and tissue repair?

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|------------------------|-------------------------|
| (A) Meiosis I | (B) Mitosis |
| (C) Reduction division | (D) Binary fission only |

Q.15 How many haploid daughter cells are produced at the completion of meiosis, and what is their chromosome count relative to the parent cell?

- | | |
|---|---|
| (A) Two cells with identical chromosome count | (B) Four cells with half the chromosome count |
| (C) Four cells with double the chromosome count | (D) Two cells with half the chromosome count |

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): Plant cells can withstand much greater osmotic changes in surrounding hypotonic solutions than animal cells.

Reason (R): Plant cells possess a rigid cellulose cell wall that exerts inward pressure against the swollen cytoplasm.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.17 Assertion (A): Mitochondria are termed the powerhouses of the cell.

Reason (R): Mitochondria produce cellular energy in the form of Adenosine Triphosphate (ATP) molecules.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.18 Assertion (A): Ribosomes are known as protein factories of the cell.

Reason (R): Ribosomes are surrounded by a double-layered lipid membrane to safeguard newly formed enzymes.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.19 Assertion (A): Chromosomes are visible as distinct rod-shaped structures only when a cell is actively dividing.

Reason (R): In a non-dividing cell, DNA remains entangled as a thread-like mass called chromatin.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.20 Assertion (A): A prokaryotic cell has a well-defined membrane-bound nucleus containing multiple linear chromosomes.

Reason (R): Bacteria and blue-green algae are prokaryotic organisms.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Osmosis Experiment with Dried Raisins and Peeled Potato

In a laboratory practical, Aryan placed 10 g of dried, shriveled raisins into a beaker of pure distilled water (Setup 1). At the same time, his classmate Priya placed swollen raisins into a concentrated 20% sodium chloride (salt) solution (Setup 2). After 4 hours, Aryan observed that his raisins had swollen significantly and became firm, while Priya observed that her raisins shrank and lost volume.

Q.21 The swelling of raisins in pure distilled water observed by Aryan is due to:

- (A) Exosmosis
(B) Endosmosis
(C) Plasmolysis
(D) Active transport

Q.22 In Setup 2, why did the raisins shrink when placed in the 20% salt solution?

- (A) Salt moved into the raisins by diffusion
(B) Water moved out of the raisins because the outer medium was hypertonic
(C) The raisin skins dissolved in salt
(D) No water movement took place

Q.23 Which natural barrier in the raisin cells acted as the semi-permeable membrane during this experiment?

- (A) The cell wall only
(B) The plasma membrane / cell membrane
(C) The nuclear envelope
(D) The vacuolar tonoplast exclusively

Q.24 If boiled raisins were used in Setup 1 instead of raw raisins, what would happen?

- (A) They would swell twice as much
(B) They would not show osmosis because boiling destroys the living cell membranes
(C) They would turn into gas
(D) They would divide rapidly

Q.25 What would happen if the swollen raisins from Setup 1 were transferred into a solution with identical water concentration (isotonic)?

- (A) They would burst instantaneously
(B) There would be no net movement of water, and raisin size would remain constant
(C) They would double in size
(D) They would immediately dissolve

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	Anton van Leeuwenhoek (1674) observed living cells in pond water. Robert Hooke (1665) had observed dead cork cells.
Q.2	(C)	Rudolf Virchow (1855) added the crucial tenet that all cells arise from pre-existing living cells by division.
Q.3	(B)	The cell membrane regulates transport, allowing essential nutrients and water to enter while blocking harmful or unwanted substances.
Q.4	(B)	In a hypotonic solution (lower solute concentration outside), water enters the cell via endosmosis. Lacking a rigid cell wall, animal cells swell and can burst (lysis).
Q.5	(B)	Plasmolysis is the withdrawal of protoplasm from the cell wall when a plant cell loses water in a hypertonic medium.
Q.6	(B)	Smooth ER synthesizes lipids and steroids, and plays a major detoxifying role in liver cells. Rough ER synthesizes proteins.
Q.7	(B)	Camillo Golgi described the Golgi apparatus, which modifies, packages, and routes proteins and complex biochemicals.
Q.8	(B)	Lysosomes contain strong hydrolytic digestive enzymes. If the cell undergoes severe damage, lysosomes rupture, causing autolysis (self-digestion).
Q.9	(A)	Mitochondria and plastids (chloroplasts) are semi-autonomous organelles with their own circular DNA and ribosomes.
Q.10	(B)	Folds (cristae) provide a huge surface area for enzymes associated with the electron transport chain and ATP synthesis.
Q.11	(C)	Leucoplasts are non-pigmented plastids whose primary function is nutrient storage (starch in amyloplasts, oils in elaioplasts).
Q.12	(B)	Cellulose provides mechanical tensile strength and structural support to plant cell walls. (Chitin is found in fungal cell walls).
Q.13	(B)	The central vacuole occupies 50-90% of the volume of mature plant cells, exerting turgor pressure against the cell wall.
Q.14	(B)	Mitosis is equational division where chromosome number is conserved, resulting in two identical daughter cells for somatic growth and repair.
Q.15	(B)	Meiosis produces 4 gamete cells, each carrying half (n) the original chromosome number of the diploid ($2n$) parent cell.
Q.16	(A)	The strong cell wall exerts wall pressure equal and opposite to turgor pressure, preventing plant cells from bursting in hypotonic media.
Q.17	(A)	Mitochondria carry out cellular respiration, synthesizing ATP which fuels virtually all endergonic biochemical processes.
Q.18	(C)	Assertion is true (ribosomes translate mRNA into proteins). Reason is false because ribosomes are non-membrane-bound ribonucleoprotein complexes.
Q.19	(A)	Chromatin condenses into tightly coiled discrete chromosomes only during prophase of cell division.
Q.20	(D)	Assertion is false because prokaryotes lack a nuclear membrane and contain an undefined nucleoid. Reason is true.
Q.21	(B)	Distilled water is hypotonic relative to raisin interior; water molecules enter raisins by endosmosis.
Q.22	(B)	The concentrated salt solution has lower water chemical potential (hypertonic), causing water to flow out (exosmosis).

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.23	(B)	The selectively permeable plasma membrane regulates water movement via osmosis.
Q.24	(B)	Boiling denatures proteins and destroys membrane integrity, rendering the cell non-living and unable to exhibit selective osmosis.
Q.25	(B)	In an isotonic medium, the rate of water entering equals the rate of water leaving; net osmotic change is zero.